Michael J. Mitchell, Ph.D.

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SUMMARY

- Enterprise data platform architect with 20 years spanning academic security research, startups, and enterprise engineering.
- Builds governed, contract-driven, auditable platforms: semantic layers, identity and access control, observability, and configuration-as-code provisioning.
- Practices agentic engineering in production: agent skill harnesses and AI-assisted review and generation workflows held to the same gates as ordinary code.
- Named inventor on 35 granted US patents with 16 applications pending; 10+ peer-reviewed publications; dozens of open-source projects.

EXPERIENCE

Enterprise Semantic Layer Architect (MTS, Solutions Architecture), T-Mobile. Bellevue, WA. *Jan 2025 - present*

- Architect of the Enterprise Semantic Layer on Microsoft Fabric, centralizing governed metrics across business domains sourced from Snowflake and Databricks.
- Turned subdomain onboarding into configuration as code: a YAML registry and Terraform provision Fabric workspaces, Entra security groups, and Azure DevOps repositories, with drift detection and audit.
- Designed the composite-group row-level-security pattern that keeps declarative nested group membership durable under automated reconciliation; enforced least-privilege, fail-closed access.
- Built the semantic layer security console and access catalog (fleet dashboard, workspace inspector, self-service access views) on Kubernetes.
- Led data-contract-driven automation on the Open Data Contract Standard, treating contracts as SLOs for observability; filed a patent on LLM-generated data contracts with automated sensitivity classification.
- Modernized lakehouse telemetry, monitoring, and alerting in Grafana; automated capex allocation from ticket and git signals.
- Built and maintain a portable agent skill harness used across Claude Code, Codex, and Cursor, plus agentic pull-request review that enforces engineering standards.

Founder, Mitchtech Labs LLC. Seattle Eastside, WA. *Nov 2023 - Jan 2025*

- Independent R&D and consulting across AI/ML, observability, and cloud infrastructure.

MTS, Software Engineer, CX Lab, T-Mobile. Bellevue, WA. *Jul 2019 - Nov 2023*

- Co-founded and directed R&D for the Customer Experience (CX) Lab's developers and contractors.
- Developed dozens of solutions and prototypes across AI/ML, AR/VR, Web3, security, and SRE domains; brought early generative AI work to executive stakeholders.
- Centralized and optimized telemetry collection, aggregation, visualization, monitoring, and anomaly detection.
- Developed and collaborated on corporate-wide standards, initiatives, review boards, and training modules.
- Developer relations: tech talks and instruction on AI, DevOps, Docker, and telemetry.

Principal Software Engineer, CX Lab, T-Mobile. Bellevue, WA. *Sep 2017 - Jul 2019*

- Directed day-to-day development, operations, and testing for an on-shore/off-shore team of developers and contractors.
- Designed, developed, and deployed Device Matrix, a remote mobile device access and CI/CD/CT platform (mission-critical for remote support during COVID-19).
- Replaced multiple third-party contracts with open-source and off-the-shelf software; annual savings of $1M+.
- Built a telemetry platform collecting millions of daily user-experience data points.

Senior Software Engineer, QoE Lab / Product Realization, T-Mobile. Bellevue, WA. *Sep 2015 - Sep 2017*

- Designed and deployed a mobile device security and CPNI/PII leak detection test suite.
- Prototyped mobile app static analysis and privacy leak detection via device logs and network capture.
- Prototyped ad-hoc synthetic user-experience testing for massive public gatherings (MUPPET).
- Applied indoor/outdoor context detection to enrich radio telemetry and diagnostic data.

Graduate Researcher, Computer Science, Florida State University. Tallahassee, FL. *2009 - 2015*

- Directed research of 8-10 students annually in security, privacy, operating systems, and storage.

- Designed and implemented Cashtags, a mobile system that protects the input and display of sensitive data (USENIX Security 2015).
- Ran a three-month human-subjects tablet usage and privacy monitoring study; co-designed malware detection for Android vendor customizations.

Founder / Software Developer, Mitchtech, LLC. Tallahassee, FL. *2008 - 2015*

- Developed a dozen+ Android applications with 500,000+ downloads; led a team of Linux developers; consulted on web, networking, and systems administration.

Graduate TA / Instructor, Computer Science, Florida State University. *2009 - 2012* - Taught kernel/device drivers, Android programming, object-oriented C++, and business applications.

Earlier: Software Developer, TextPromoter (dba Catchwind Mobile Marketing), Des Moines, IA (2005 - 2007). Network Administration Intern, Briggs & Stratton, Murray, KY (2004).

EDUCATION

- **Ph.D. Computer Science**, Florida State University, 2015. Dissertation: *Cashtags: Protecting the Input and Display of Sensitive Data*.
- **M.S. Computer Science**, Florida State University, 2011.
- Computer Science / Management Information Systems coursework, Iowa State University, 2002 - 2007.

SELECTED PATENTS

Full list of 35 granted patents: michaeljmitchell.com/patents

- Fine-Tuning Images Generated by Artificial Intelligence Based on Aesthetic and Accuracy Metrics (US 12,682,506)
- Caching System for Presenting Dashboard Snapshots with Quick Response Access to Live Dashboard Data (US 12,326,912)
- Generating Encoded Text Based on Spoken Utterances Using Machine Learning (US 11,880,645; US 12,248,748)
- Using Machine Learning to Optimize Wireless Carrier System Testing (US 11,812,290)
- Honeypot Adaptive Security System (US 11,075,946)
- eSIM Switching Platform (US 11,129,026)
- Echolocate Spectrum Analyzer (US 10,827,168)
- Distributed Testing of Wireless Networks (US 10,382,922)
- Voice User Interface for Data Access Control (US 10,496,363)
- Obfuscation of Sensitive Human-Perceptual Output (US 10,025,952)

SELECTED PUBLICATIONS

- **Cashtags: Protecting the Input and Display of Sensitive Data.** M. Mitchell, A. Wang, P. Reiher. *USENIX Security Symposium*, 2015.
- **Mobile Usage Patterns and Privacy Implications.** M. Mitchell et al. *IEEE PerMoby*, 2015.
- **Systematic Audit of Third-Party Android Phones.** M. Mitchell, G. Tian, Z. Wang. *ACM CODASPY*, 2014.
- **TrueErase: Per-file Secure Deletion for the Storage Data Path.** S. Diesburg et al. *ACSAC*, 2012.
- **ContextProvider: Context-Awareness for Medical Monitoring Applications.** M. Mitchell et al. *IEEE EMBC*, 2011.

CERTIFICATIONS AND AWARDS

- FAA Part 107 Remote Pilot. CISSP, 2018.
- Elasticsearch Business Transformation Award, 2019. T-Mobile Enterprise IT Self Service VIP, 2016.
- NSTISSI-4011 and CNSSI-4014 information assurance certifications.
- Winner, ACM D-FENSE offensive security competition, 2013. Top Accelerator Team, Microsoft Imagine Cup, 2013.
- Runner-up, FSU 3 Minute Thesis, 2014. Eagle Scout, BSA.